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CHAPTER 20

Nuclear Power Reactors

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We concentrate in this chapter on existing and some future power reactors and discuss the features common to each generic type as well as some peculiarities of special designs. Waste generation and handling at the power plants is also described generically.

Electric power producing reactors have been built to 1500 MW_e but all sizes in the range 400−1400 MW_e are common. With increasing size the cost of the electric power usually decreases. Power reactors of ~200 MW_e are competitive in some areas of high energy costs, such as in many developing countries. These smaller sized plants have some advantages: they are more reliable due to less stress of the components, they can be built more easily in remote areas because the smaller components are easier to transport, the electric grid may be more capable of accepting the moderate power addition, and they can more easily be designed with inherent safety features. Table 20.1 summarizes the characteristics of some operating nuclear power reactors.

In an attempt to systematise the continuous development of nuclear power reactors the concept of reactor generations has been introduced. In 2012 about 4 generations are discussed. The third has actually been divided into two different ones due to the clear trend break for the last generation where the novelty is not only the reactor but a coupled, optimised system of techniques.

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